

Aadesh Salecha

MS @ STANFORD | AI/ML @ AMAZON | EX-DRAFTKINGS | MULTI-PUBLISHED RESEARCHER

☎ 612-391-6966 | ✉ asalecha@stanford.edu | 🏠 bit.ly/ASGoogleScholar | 📷 [aadeshsalecha](#) | 🌐 [aadesh-salecha](#)

Education

Masters of Science: Computer Science

STANFORD UNIVERSITY

Sep 2024 - June 2026

Primary Advisor: Prof. Johannes Eichstaedt

Bachelor of Science: Computer Science

UNIVERSITY OF MINNESOTA

Twin Cities

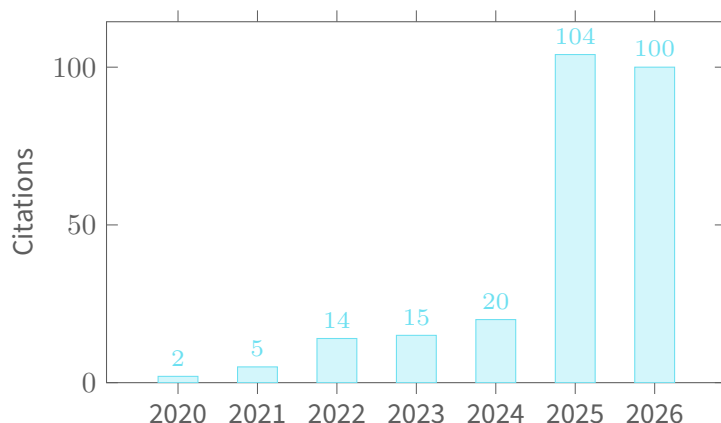
Aug 2018 - May 2021

Honors: *Summa Cum Laude*

GPA: 4.0/4.0

Publications

HIGHLIGHTS



Total Publications: 16
Total citations: 178
h-index: 6

1. **Large Language Models Display Human-like Social Desirability Biases in Big Five Personality Surveys**
Aadesh Salecha, Molly E. Ireland, Shashanka Subrahmanya, João Sedoc, Lyle H. Ungar, and Johannes C. Eichstaedt. *PNAS Nexus*, Vol. 3, No. 12, pg. e533, 2024. academic.oup.com/pnasnexus/article/3/12/pgae533/7919163
[WIRED Magazine Article: Chatbots, Like the Rest of Us, Just Want to Be Loved](#)

JOURNAL PUBLICATIONS

2. **Monitoring the Opioid Epidemic via Social Media Discussions.**
Adam Lavertu, Tymor Hamamsy, Aadesh Salecha, Delaney Smith, Johannes Eichstaedt, and Russ B Altman. *Nature Partner Journals: Digital Medicine*. nature.com/articles/s41746-025-01642-x
3. **Fake News Spreader Detection using Trust-based strategies in Social Networks with Bot Filtration.**
Bhavtosh Rath, Aadesh Salecha, and Jaideep Srivastava. *Social Network Analysis and Mining (SNAM)* (2022)
doi.org/10.1007/s13278-022-00890-z
4. **Relationship between Citizen-Eyewitness Images and Audience Engagement with News.**
Jisu Kim, Jisu Huh, Bhavtosh Rath, Aadesh Salecha, and Jaideep Srivastava. *Journalism Practice* (2021)
doi.org/10.1080/17512786.2021.1882873
5. **Influence of Emotions on Coping Behaviors in Crisis: A Computational Analysis of the COVID-19 Outbreak.**
Hao Xu, Smitha Muthya Sudheendra, Jisu Huh, Aadesh Salecha, and Jaideep Srivastava. *Journal of Computational Social Science*, 7, 1599–1623 (2024). <https://doi.org/10.1007/s42001-024-00282-7>

6. **The Language of Conflict Transformation - Assessing Psychological Change Patterns in Israeli-Palestinian Track Two Interactive Problem Solving.**
Oliver Fink, Wilfried Graf, Shashanka Subrahmanya, **Aadesh Salecha**, and Johannes Eichstaedt. *Negotiation and Conflict Management Research* (2024)
7. **TherapyTrainer: Using AI to Train Therapists in Written Exposure Therapy.**
Elizabeth C. Stade, Johannes C. Eichstaedt, Debra L. Kaysen, **Aadesh Salecha**, Alanna Greenberger, Shreya Singhvi, and Shannon Wiltsey Stirman. *Cognitive and Behavioral Practice* (2025)
8. **No Good Time to Negotiate? Effective Track Two Dialogue in Protracted Israel-Palestine Conflict Escalation.**
Oliver Fink, Wilfried Graf, Shashanka Subrahmanya, **Aadesh Salecha**, and Johannes Eichstaedt. *Conflict Resolution Quarterly* (2026)
doi.org/10.1002/crq.70032

CONFERENCE PUBLICATIONS

9. **Emergence and Stability of Self-Evolved Cooperative Strategies using Stochastic Machines.**
Jin Hong Kuan, and **Aadesh Salecha**. In *IEEE Symposium Series on Computational Intelligence (SSCI)* (2020)
<https://ieeexplore.ieee.org/abstract/document/9308258>
10. **Detecting Fake News Spreaders in Social Networks using Inductive Representation Learning.**
Bhavtosh Rath, **Aadesh Salecha**, and Jaideep Srivastava. In *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)* (2020) <https://ieeexplore.ieee.org/abstract/document/9381466>
11. **Public's Discrete Emotions During the COVID-19 and Impacts on Their Coping Behaviors in the U.S.**
Hao Xu, Smitha Muthya Sudheendra, Jisu Huh, **Aadesh Salecha**, and Jaideep Srivastava. *72nd Annual International Communication Association Conference* (2022)

UNDER REVIEW

12. **A Computational Model of Reward Learning and Habits on Social Media**
Georgia Turner, Lukas J. Gunschera, Shashanka Subrahmanya, **Aadesh Salecha**, Johannes C. Eichstaedt, Stefano Palminteri, and Amy Orben. Accepted, *Nature Communications*. osf.io/preprints/psyarxiv/xe25k
13. **The answer to the mental health crisis: Examining media bias regarding psychedelic treatments for depression and PTSD using natural language processing.**
Audrey Evers, Elizabeth C. Stade, **Aadesh Salecha**, Zoe Tait, Gabriela Khazanov.
14. **Structured AI Dialogues Can Increase Happiness and Meaning in Life**
Aadesh Salecha, Jonas Schöne, Sonja Lyubomirsky, João Sedoc, Johannes Eichstaedt, and Robb Willer.
15. **Increased Behavioural Sensitivity to 'Likes' During the COVID-19 Pandemic.**
Brandon C. Davidson, **Aadesh Salecha**, et al. Submitted to *Psychological Science* (2026)

THESES

16. **Empirical Study of the Spread of Misinformation: A Big Data Approach.**
Aadesh Salecha. Undergraduate Thesis (Summa Cum Laude). Retrieved from the *University of Minnesota Digital Conservancy* (2021). <https://hdl.handle.net/11299/222250>

Press Coverage

1. **Chatbots, Like the Rest of Us, Just Want to Be Loved.**
Wired . Published on March 5, 2025. Retrieved from <https://www.wired.com/story/chatbots-like-the-rest-of-us-just-want-to-be-loved/>
2. **Breaking Negative Thought Patterns Could Ward Off Anxiety, Depression.**
Science News . Published in February 2025. Retrieved from <https://www.sciencenews.org/article/negative-thoughts-anxiety-depression>

3. **Are AI Chatbot 'Personalities' in the Eye of the Beholder?**

Science News . Published in February 2025. Retrieved from <https://www.sciencenews.org/article/ai-chatbot-personalities>

4. **Large Language Models Just Want to Be Liked.**

Stanford HAI . Published on January 13, 2025. Retrieved from <https://hai.stanford.edu/news/large-language-models-just-want-be-liked>

Academic Work Experience

Computational Psychology & Well-Being Lab

Stanford University

AI/ML SCIENTIST

May 2023 – Present

Technologies: *Differential Language Analysis, Large Language Models*

Mentored by *Prof. Johannes Eichstaedt*

- Developing **LLM evaluation metrics** to measure **behavioral** strengths and **biases** based on psychological principles^[8]
- Creating **AI-assisted** variants of well-established and novel positive-psychology **interventions** to improve well-being^[11]
- Applying advanced **NLP** techniques to novel domains like **conflict mediation** talks between **Israel & Palestine**^[9,10]
- Facilitating measurement and causal analysis of **population health metrics like opioid usage**^[4] and subjective well-being

Data Analysis and Management Research Group

University of Minnesota

RESEARCH ASSISTANT

Feb 2019 – Dec 2022

Technologies: *Graph Neural Networks, NoSQL*

Mentored by *Prof. Jaideep Srivastava*

- Curated a dataset of 10 unique news events and 50 million associated users and their follower-followee networks^[6]
- Published methods that leverage inter-user trust to **detect fake news spreaders** with an accuracy of 93%^[1]
- Analyzed people's information seeking/sharing behavior in relation to COVID-19 social-distancing compliance^[3,7]

Industry Experience

Amazon (AWS)

Seattle

AI/ML SOFTWARE ENGINEER

June 2025 – Sept 2025

Technologies: *Large Language Models, RAG, Agentic AI, Prometheus, Grafana, CloudWatch*

- Designed an **agentic AI system** leveraging **RAG** to autonomously diagnose and resolve ML training failures across distributed GPU clusters
- Integrated telemetry data (Prometheus, Grafana, CloudWatch) into a data-model feedback loop for **predictive maintenance** and system reliability
- Improved compute efficiency and reduced manual effort, cutting operational costs and lowering **carbon emissions by 200 kg CO₂e per day**

DraftKings Inc.

Boston

SENIOR SOFTWARE ENGINEER

Dec 2021 – Jun 2024

Technologies: *.NET, MySQL, AWS, RabbitMQ, Octopus Deploy*

- Co-developed a **distributed accounting ledger system** to store millions of user transactions
- Delivered **full-stack** solutions, micro-services to handle billions of requests with <200ms latencies
- **Led cross-functional** initiatives to elevate user experience and reduce dropout by 40%

Blue Zones, LLC

Remote

DATA SCIENTIST

Jun 2023 - Jan 2024

Worked with *Dan Buettner*, New York Times best-selling author

- Curated a novel **dataset of over 1 million recipes**, analyzing American culinary preferences
- Leveraged advanced *data mining techniques* to pinpoint recipe candidates for Dan Buettner's upcoming book
- Implemented a **production-grade** recipe generation bot using **RAG, and advanced prompt engineering** techniques

Awards & Honors

2020	Undergraduate Research Opportunity Grant , Office of Research Opportunities	\$3000.00
	Tom Niccum Travel Grant , College of Science and Engineering	\$1000.00
2019	Google's CS Research Mentorship Program Nominee , Google, Inc.	
	Dean's List - Spring 2019 , College of Science and Engineering	
	Dean's List - Fall 2019 , College of Science and Engineering	
2018	ExxonMobil Scholarship , ExxonMobil via Computer Science Department, UMN	\$3000.00
	Dean's List - Fall 2018 , College of Science and Engineering	

Conference Presentations

Kuan, Jin Hong, and **Aadesh Salecha**. "Emergence and Stability of Self-Evolved Cooperative Strategies using Stochastic Machines." *IEEE Symposium Series on Computational Intelligence (SSCI)*, Canberra, Australia. (2020)

Teaching Experience

Average Student Rating: 4.75/5, Total Teaching Hours: 900+ hours

Winter '25	OIT 274: Data and Decisions, Teaching Assistant with Prof. Claudia Santa Cruz
Autumn '25	OIT 247: Optimization and Simulation Modeling - Accelerated, Teaching Assistant with Prof. Kostas Bimpikis
Spring '21	CSCI 2033: Computational Linear Algebra, Teaching Assistant with Prof. Carl Sturtivant
Summer '20	CSCI 1933: Intro to DS & Algo, Teaching Assistant with Prof. James Polzin
Fall '20	CSCI 2041: Advanced Programming Principles, Teaching Assistant with Prof. Eric Van Wyk
Spring '20	CSCI 2011: Discrete Mathematics, Teaching Assistant with Prof. Kate Jensen
Spring '20 – '21	Computer Science, Mathematics, and Physics, Peer Tutor at SMART Learning Commons

Academic Service

2022	Committee Member , 16th International Conference on Advanced Engineering Computing and Applications in Sciences (ADVCOMP)
2020 - 2021	Peer Reviewer , Minnesota Undergraduate Research & Academic Journal (MURAJ) Peer Reviewed 2 articles published in Volume 4, Issue 5 of the MURAJ Undergraduate Journal